

5G RAN

Stable Video Surveillance Experience Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.
Added the impact analysis of avoidance of RedCap uplink scheduling. For details, see [5.2.2 Impacts](#).

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-06-28).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Enhanced the function of avoidance of RedCap uplink scheduling in NR TDD. For details, see: • 5.1 Principles • 5.4.1 Data Configuration	Added the NRDUCellRedCap.UlAvoidSinrCoeff parameter. Modified parameter: Added the enumerated value LEVEL3 to the NRDUCellRedCap.UlAvoidPol parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-080011	Stable Video Surveillance Experience	4 Stable Video Surveillance Experience 5 Avoidance of RedCap Uplink Scheduling

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Stable Video Surveillance Experience	None	4 Stable Video Surveillance Experience
Avoidance of RedCap uplink scheduling	Supported only in SA networking	5 Avoidance of RedCap Uplink Scheduling

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Stable Video Surveillance Experience	Supported only in low frequency bands	4 Stable Video Surveillance Experience 5 Avoidance of RedCap Uplink Scheduling

3 Overview

In scenarios with large volume of video surveillance services, such as ports, steels, and coal mines, as the number of UEs requiring rate guarantee increases, video frame freezing occurs, deteriorating user experience. To ensure the uplink transmission rate and the number of UE connections, and improve the stability of video surveillance services, Huawei introduces the Stable Video Surveillance Experience feature. Typical video surveillance services consist of I-frames and P-frames.

- I-frames are key frames of images and contain complete images.
- P-frames are prediction frames. P-frames are decoded based on key frames to generate final images.

4 Stable Video Surveillance Experience

4.1 Principles

The Stable Video Surveillance Experience function is controlled by the **STABLE_UL_SERVICE_EXP_BASIC_SW** option of the **NRDUCellFeatureSw.StableULServiceExpSwitch** parameter. **After this function is enabled, the gNodeB guarantees the uplink transmission rate of UEs based on the queuing mechanisms, scheduling priorities, and rank algorithms to improve user experience of video surveillance services.**

This function is applicable to non-RedCap UEs and NR TDD RedCap UEs.

Stable Video Surveillance Experience takes effect at the UE level. Therefore, the following configurations are required:

- Set the **gNBDUMacParamGroup.SchedulingType** parameter to **STABLE_UL_SERVICE_EXP_SCH** to enable Stable Video Surveillance Experience for UEs with QCI-specific bearers.
- Set the **gNBQciBearer.ServiceType** parameter to **Normal**.
- Set the **NRDUCellUISch.StableUIExpMaxUeNum** parameter based on the required number of UEs.
- For video surveillance UEs, set the **gNBDUSchParamGroup.StableUIExpGuarRate** parameter based on the uplink rate requirements of UEs.
- For RedCap UEs performing video services, set the **gNBDUSchParamGroup.RedCapStableUIExpGuarRate** parameter based on the uplink rate requirements of UEs.
- Set the **gNBDUSchParamGroup.StableUIExpPol** parameter based on different service requirements of UEs.
 - When the number of UEs is small and the UEs have high requirements on video services, set this parameter to **EXPERIENCE_FIRST** to achieve the optimal guaranteed rate for UEs.
 - If there are a large number of UEs and the instantaneous rate of more UEs is expected to reach the guaranteed rate, set this parameter to **USER_NUM_FIRST** to increase the number of UEs that meet related QoS requirements.

- If both the UE rate and the number of UEs meeting QoS requirements need to be guaranteed, set this parameter to **NEUTRAL**.

Stable Video Surveillance Experience involves the following:

- Queuing mechanism: UEs performing video surveillance services in a cell are classified into two types: UEs that meet QoS requirements and UEs that do not meet QoS requirements.
- Scheduling priority: UEs that do not meet QoS requirements are preferentially scheduled so that more video surveillance service UEs can reach the guaranteed uplink rate for stable uplink service experience. If the QoS requirements are met, no special processing is required.
- Rank algorithm: The rank is lowered for UEs that meet QoS requirements so that more space-domain resources can be saved for UEs that do not meet QoS requirements. In this way, more video surveillance UEs can obtain space-domain resources and reach the guaranteed uplink rate for stable uplink service experience.

NOTE

When the instantaneous rate of a UE is greater than or equal to the guaranteed uplink rate for stable uplink service experience, the UE is regarded as a UE that meets the QoS requirements.

When the instantaneous rate of a UE is lower than the guaranteed uplink rate for stable uplink service experience, the UE is regarded as a UE that does not meet the QoS requirements.

After the QoS requirements of a UE are met, if the uplink traffic volume of the UE is large, the base station performs scheduling guarantee based on the traffic volume. Therefore, the uplink rate of the UE will exceed the guaranteed uplink rate.

Only one stable uplink service experience bearer can be configured for a UE to be enabled with stable uplink service experience. If more than one such bearer is configured for the UE, this function does not take effect.

NOTE

This feature is a commercial feature in NR TDD.

4.2 Network Analysis

4.2.1 Benefits

When all the following conditions are met, the number of UEs whose video experience is guaranteed increases by 10% to 30% in NR TDD. The gains depend on actual services. If the following conditions are not met, the gains provided by this feature decrease or are not significant:

Table 4-1 Conditions for Stable Video Surveillance Experience gains

UE Requirement	• UEs are evenly distributed at the cell center (RSRP greater than -75 dBm), a medium distance from the cell center (RSRP ranging from -85 dBm to -75 dBm), and cell edge (RSRP ranging from -95 dBm to -85 dBm).
-----------------------	---

Cell Requirement	- The cell capacity is limited (that is, the total rate required by video surveillance UEs exceeds the available bandwidth capability), and the number of cameras with smooth videos is less than the number of cameras that can be accommodated in the cell (depending on the camera location and service model). - Uplink MU-MIMO is enabled. - The gNBDUSchParamGroup.StableUIExpPol parameter is set to USER_NUM_FIRST.
Service Requirement	- UEs with the same type of QCI connect to the same number of cameras. - The incoming packet model of UEs is a camera surveillance or remote control video service. The peak-to-average ratio (PAR) of I frames is set to 20, and the buffer is set based on the actual camera type.

NOTE

The **gNBDUSchParamGroup.StableUIExpGuarRate** parameter is set based on the following formula:

- I-frame size (Mbit) = (Video bit rate / ((FPS - 1) + PAR)) × PAR × 1s
- P-frame size (Mbit) = (Video bit rate - I-frame size) / (FPS - 1)
- Guaranteed uplink rate for stable uplink service experience (Mbit/s) = (I-frame size + (Number of cameras connected to the UE - 1) × P-frame size) / Camera buffer duration

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

After the Stable Video Surveillance Experience function takes effect, as UEs whose data rates meet QoS requirements occupy a large number of uplink resources, resources available to other UEs decrease, the average uplink cell throughput indicated by **Cell Uplink Average Throughput (DU)** may decrease and service experience of cameras with poor coverage at some locations deteriorates.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-080011	Stable Video Surveillance Experience	NR0S00SVSE00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapBasedFunSw parameter	<i>RedCap</i>	For NR TDD RedCap UEs, stable uplink service experience for RedCap UEs takes effect only when RedCap Introduction is enabled.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	None	None	None	None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	VoNR	VONR_SW option of the NRCellAlgoSwitch . <i>VonrSwitch</i> parameter	<i>VoNR</i>	The Stable Video Surveillance Experience function does not take effect for UEs for which VoNR has taken effect.
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch . <i>HighReliabilitySwitch</i> parameter	<i>URLLC</i>	The Stable Video Surveillance Experience function does not take effect for UEs with low latency and high reliability enabled.
Low-frequency TDD	Uplink live webcast experience guarantee	UL_LIVE_WEBCAST_EXP_GUAR_SW option of the NRDUCellFeatureSw . <i>QciBasedDiffServExpSw</i> parameter	<i>Ultimate Video Experience</i>	When both uplink live webcast experience guarantee and Stable Video Experience Surveillance are enabled, the number of video surveillance UEs for which stable uplink service experience is guaranteed may be affected.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-2 describes the parameters used for function activation.

Table 4-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Stable Uplink Service Experience Switch	NRDUCellFeatureSw. <i>StableUlServiceExpSwitch</i>	Select the STABLE_UL_SERVICE_EXP_BASIC_SW option if Stable Video Surveillance Experience is required.
Scheduling Type	gNBDUMacParamGroup. <i>SchedulingType</i>	Set this parameter to STABLE_UL_SERVICE_EXP_SCH if Stable Video Surveillance Experience is required.
Service Type	gNBQciBearer. <i>ServiceType</i>	Set this parameter to NORMAL if Stable Video Surveillance Experience is required.

Parameter Name	Parameter ID	Setting Notes
Stable Uplink Experience Guaranteed Rate	gNBDUSchParamGroup.StableULExpGuarRate	Set this parameter based on the uplink rate requirements of UEs. To achieve the optimal gains, set this parameter by referring to the method of calculating the guaranteed rate in 4.2.1 Benefits. • If this parameter is set to an excessively large value, the actual service rate cannot reach the configured guaranteed rate, and the expected gains cannot be achieved. • If this parameter is set to an excessively small value, the actual service rate can reach the configured guaranteed rate, but video surveillance services may still experience frame freezing.
RedCap Stable UL Exp Guaranteed Rate	gNBDUSchParamGroup.RedCapStableULExpGuarRate	Set this parameter based on the uplink rate requirements of UEs. To achieve the optimal gains, configure this parameter by referring to the calculation method of the guaranteed rate in 4.2.1 Benefits. • If this parameter is set to an excessively large value, the actual service rate cannot reach the configured guaranteed rate, and the expected gains cannot be achieved. • If this parameter is set to an excessively small value, the actual service rate can reach the configured guaranteed rate, but video surveillance services may still experience frame freezing.

Parameter Name	Parameter ID	Setting Notes
Stable Uplink Experience Policy	gNBDUSchParamGroup.StableULExpPol	Set this parameter based on customer policies: • If the gNodeB needs to preferentially guarantee the rates of UEs for which stable uplink service experience takes effect, set this parameter to EXPERIENCE_FIRST. • If the gNodeB needs to preferentially guarantee the capacity in terms of the number of UEs for which stable uplink service experience takes effect, set this parameter to USER_NUM_FIRST. • If both the rates and the capacity in terms of the number of UEs for which stable uplink service experience takes effect need to be guaranteed, set this parameter to NEUTRAL.
Stable UL Service Exp Max UE Num	NRDUCellUISch.StableULExpMaxUeNum	Configure this parameter based on the required number of UEs. If the NRDUCellUISch.StableULExpMaxUeNum parameter is set to a value other than 255, it is recommended that the NRDUCellQciBearer.UeInactivityTimer and NRDUCellQciBearer.RedCapUeInactivityTimer parameters be set to 10s to prevent cameras with no services from occupying the specifications used for guarantee.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling Stable Video Surveillance Experience
MOD NRDUCELLFEATURESW: NrDuCellId=1,
StableUIServiceExpSwitch=STABLE_UL_SERVICE_EXP_BASIC_SW-1;

//Adding a bearer for video surveillance services (If a single bearer with the 5QI of 82 needs to be set up for
a UE, the core network must support the bearer.)
ADD GNB5QICONFIG: Nr5qi=82, Qci=82;
ADD GNBQCIBEARER: Qci=82;
//For all cells served by the base station, if the bearer corresponding to a QCI needs to be configured for
video surveillance services, the service type must be set to common service.
MOD GNBQCIBEARER: Qci=82, ServiceType=NORMAL;

//Configuring the video surveillance services: setting the bearer for video surveillance services, and
associating the bearer with the QCI (Configurations for an SUL cell must be set independently.)
ADD NRCELLQCIBEARER: NrCellId=1, Qci=82;
ADD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82;
ADD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1;
ADD GNBUMACPARAMGROUP: MacParamGroupId=1;
MOD GNBUMACPARAMGROUP: MacParamGroupId=1, SchedulingType=STABLE_UL_SERVICE_EXP_SCH;
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82, MacParamGroupId=1, ScheduleParamGroupId=1;

//Configuring the maximum number of UEs with stable uplink service experience
MOD NRDUCELLULSCH: NrDuCellId=1, StableUIExpMaxUeNum=10;

//Configuring the guaranteed rate and policy for Stable Video Surveillance Experience when both the rates
and the capacity in terms of the number of UEs for which stable uplink service experience takes effect need
to be considered (The value 2000 of gNBUSchParamGroup.StableUIExpGuarRate indicates 2 Mbit/s)
MOD GNBUSCHPARAMGROUP: ScheduleParamGroupId=1, StableUIExpGuarRate=2000,
StableUIExpPol=USER_NUM_FIRST;

//Triggering RRC connection reconfiguration after online UEs modify parameters related to Stable Video
Surveillance Experience (that is, after the bearer and corresponding QCI configurations for video
surveillance services are modified)
STR NRCELLSERVICEPARAMEFFECT: NrCellId=1, NrNetworkingOption=SA, First5qi=82;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling Stable Video Surveillance Experience
MOD NRDUCELLFEATURESW: NrDuCellId=1,
StableUIServiceExpSwitch=STABLE_UL_SERVICE_EXP_BASIC_SW-0;
```

4.4.2 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.3 Activation Verification

Observe the total traffic volume of uplink data with a specific 5QI received at the RLC layer in a DU cell (**N.ThpVol.UL.DuCell5QI**). If the value increases, this function has taken effect.

4.4.4 Network Monitoring

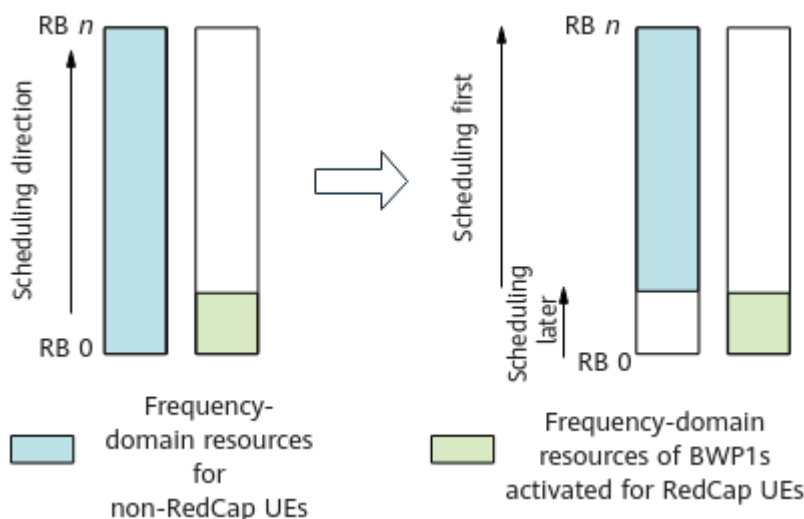
For details, see [4.2.2 Impacts](#).

5 Avoidance of RedCap Uplink Scheduling

5.1 Principles

When RedCap UEs coexist with non-RedCap high-bandwidth UEs, the avoidance of RedCap uplink scheduling function is introduced to reduce resource contention and improve video smoothness of RedCap UEs.

After this function takes effect, non-RedCap UEs determine whether to avoid the frequency-domain resources of all BWP1s activated for RedCap UEs (referred to as avoidance positions) based on the current frequency-domain SINR. That is, if the SINR meets the avoidance conditions, the gNodeB first allocates frequency-domain resources outside the avoidance positions during uplink scheduling and, if necessary, within the avoidance positions. This reduces resource contention between non-RedCap UEs and RedCap UEs and increases the average uplink throughput of RedCap UEs.



This function is controlled by the **NRDUCellRedCap.UIAvoidPol** parameter. Different avoidance policies can be selected for non-RedCap UEs with different priorities, as described in [Table 5-1](#).

Table 5-1 Policies involved in avoidance of RedCap uplink scheduling

Avoidance Policy	Avoidance Method	Parameter Setting
Policy 1	The gNodeB does not avoid the BWP frequency-domain positions corresponding to RedCap UEs during uplink scheduling for non-RedCap UEs.	NRDUCellRedCap.UIAvoidPol set to LEVEL0
Policy 2	If the SINR of non-RedCap UEs on the current frequency domain meets the avoidance conditions, non-RedCap UEs with medium and low priorities avoid the BWP frequency-domain positions corresponding to RedCap UEs during uplink scheduling.	NRDUCellRedCap.UIAvoidPol set to LEVEL1
Policy 3	If the SINR of non-RedCap UEs on the current frequency domain meets the avoidance conditions, non-RedCap UEs with medium, low, and relatively high priorities avoid the BWP frequency-domain positions corresponding to RedCap UEs during uplink scheduling.	NRDUCellRedCap.UIAvoidPol set to LEVEL2
Policy 4	If the SINR of non-RedCap UEs on the current frequency domain meets the avoidance conditions, non-RedCap UEs with medium, low, and relatively high priorities avoid the BWP frequency-domain positions corresponding to RedCap UEs based on the frequency-domain positions of BWP1s activated for RedCap UEs during uplink scheduling.	NRDUCellRedCap.UIAvoidPol set to LEVEL3

Whether the SINR of non-RedCap UEs on the current frequency domain meets the avoidance conditions depends on the configuration of the **NRDUCellRedCap.UIAvoidSinrCoeff** parameter. A larger value of this parameter indicates a higher probability to trigger resource avoidance.

5.2 Network Analysis

5.2.1 Benefits

This function improves the video smoothness of RedCap UEs when there are a large number of small-packet services for non-RedCap UEs. The increase depends on the scheduling priority of RedCap UEs and the proportion of MU pairing. A

lower scheduling priority of RedCap UEs and a lower proportion of MU pairing indicate a larger increase.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

The average uplink throughput of non-RedCap UEs slightly decreases because non-RedCap UEs avoid the frequency-domain resources of all BWP1s activated for RedCap UEs and the scheduling positions change. The higher the value of the **NRDUCellRedCap.UlAvoidPol** parameter, the more non-RedCap UEs are affected.

When the **NRDUCellRedCap.UlAvoidPol** parameter is set to **LEVEL1** or **LEVEL2** and the **UL_SCHEDULE_ENH_SW** option of the **NRDUCELLPUSCH.UlPuschAlgoSwitch** parameter is deselected, there are no gains for RedCap UEs in some scenarios where the PRB load is heavy.

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-080011	Stable Video Surveillance Experience	NR0S00SVSE00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	REDCAP_SW option of the NRDUCellFeatureSw.UeCapbBasedFunSw parameter	<i>RedCap</i>	The avoidance of RedCap uplink scheduling can be enabled only when RedCap Introduction is enabled.
Low-frequency TDD	Optimal paired RB selection	OPT_PAIR_RB_SEL_SW option of the NRDUCellULMimo.UlMuMimoAlgoSwitch parameter	None	When the UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter is selected, the avoidance of RedCap uplink scheduling function takes effect only when the optimal paired RB selection switch is turned on.

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	Uplink interference randomizati on-based scheduling	NRDUCellAlgoSwitch.UlIntrfRandomSchSwitch	<i>Uplink Scheduling</i>	When enabled together with avoidance of RedCap uplink scheduling, uplink interference randomization-based scheduling will produce reduced benefits for non-RedCap UEs. This is because non-RedCap UEs are scheduled from frequency-domain resources other than those on the BWP1s activated for RedCap UEs.
Lo w- freq uen cy TD D	Uplink frequency selective scheduling	None	<i>Uplink Scheduling</i>	When enabled together with avoidance of RedCap uplink scheduling, uplink frequency selective scheduling will produce reduced benefits for non-RedCap UEs. This is because non-RedCap UEs are scheduled from frequency-domain resources other than those on the BWP1s activated for RedCap UEs.

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TD D	Uplink MU grouping and pairing optimization for RedCap UEs	REDCAP_UL_MU_GRP_PAIR_OPT_SW option of the NRDUCellRedCap.RedCapSchAlgoSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	When the NRDUCellRedCap.UplinkAvoidPol parameter is set to LEVEL1 or LEVEL2 and uplink MU grouping and pairing optimization for RedCap UEs is enabled, the number of uplink RBs that can be used for avoidance in the activated BWP1s for RedCap UEs may be limited. This function may not produce gains for RedCap UEs.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Cells

The **NRDUCell.FrequencyBand** parameter must be set to **N38, N39, N40, N41, N77, N78, or N79**.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-2](#) describes the parameters used for function activation.

Table 5-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
UL Avoidance Policy	NRDUCellRedCap.UlAvoidPol	When RedCap UEs contend for resources with non-RedCap large-bandwidth UEs: • It is recommended that this parameter be set to LEVEL1 if the average uplink throughput of RedCap UEs needs to be increased (non-RedCap UEs with medium and low priorities try to avoid the BWP frequency-domain positions corresponding to RedCap UEs based on SINRs during uplink scheduling). • It is recommended that this parameter be set to LEVEL2 if the average uplink throughput of RedCap UEs needs to be further increased (non-RedCap UEs with medium, low, and relatively high priorities try to avoid the BWP frequency-domain positions corresponding to RedCap UEs based on SINRs during uplink scheduling). • In heavy-traffic scenarios, it is recommended that this parameter be set to LEVEL3 if non-RedCap UEs with medium, low, and relatively high priorities need to adaptively avoid the BWP frequency-domain positions corresponding to RedCap UEs based on SINRs and scheduling behaviors during uplink scheduling. A higher level (indicated by a larger number following LEVEL in the parameter value) results in a higher average throughput of RedCap UEs

Parameter Name	Parameter ID	Setting Notes
		and a larger possible decrease in the average throughput of non-RedCap UEs.
UL Avoidance SINR Coefficient	NRDUCellRedCap.UlAvoidSinrCoeff	Set this parameter based on the network plan.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling avoidance of RedCap uplink scheduling (LEVEL1 used as an example) and setting UL Avoidance SINR Coefficient (50 used as an example)
MOD NRDUCELLREDCAP: NrDuCellId=0, UIAvoidPol=LEVEL1, UIAvoidSinrCoeff=50;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling avoidance of RedCap uplink scheduling (LEVEL1 used as an example)
MOD NRDUCELLREDCAP: NrDuCellId=0, UIAvoidPol=LEVEL0;
```

5.4.2 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.3 Activation Verification

Observe the total traffic volume of uplink data with a specific 5QI received at the RLC layer in a DU cell (**N.ThpVol.UL.DuCell5QI**). If the value increases, this function has taken effect.

5.4.4 Network Monitoring

For details, see [5.2.2 Impacts](#).

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 References

1. 3GPP TS 38.104: "NR;Base Station (BS) radio transmission and reception"
2. *3900 & 5900 Series Base Station Product Documentation*
3. *VoNR*
4. *URLLC*
5. *RedCap*
6. *License Control Item Lists*
7. *Ultimate Video Experience*
8. *Uplink Scheduling*
9. *Massive MIMO Uplink Boosting (Low-Frequency TDD)*